

8) The Scenario

The devices have the following addresses:

Device	IP Address	Subnet Mask
Router	192.168.2.1	128
Windows PC	192.168.2.	128
Linux	192.168.2.29	128
Mac	192.168.2.129	128

a) why can windows access the internet
but Linux & Mac cannot?

Ans => Because they are not in the same subnet as the router

9) ~~what~~ does 128 mean?

11111111 . 11111111 . 11111111 . 11110000

(or)

255 . 255 . 255 . 240

Network bits = 28

Host bits = 4

In last octet :

11110000
network Host

10) why only 4 Host Bits?

IPv4 always has 32 bits

if 28 bits = Network

then $32 - 28 = 4$ bits

So the host can only use

0000 $\rightarrow$ 1111

that's

$2^4 = 16$ addresses

First address = Network Address

Last address = Broadcast address

usable host $\Rightarrow 16 - 2 \Rightarrow 14$ Hosts

11) How do we find the network address?

Rule $\Rightarrow$ Fill every host bit with 0.

e.g. Router

192.168.2.1 / 28

Host octet $\Rightarrow$ 1 1 1 1

1 = 0000 0001

Subnet mask

240 = 1111 0000

Now separate network & host

IP $\Rightarrow$ 0000 10001

Mask $\Rightarrow$ 1111 10000

* Network bits remains

* Host bits become zero.

0000 10000 $\Rightarrow$ decimal (0)

$\therefore$ Network address

192.168.2.0

12) First Host Address

Rule $\Rightarrow$ Network address + 1

192.168.2.1

6) Broadcast address

Rule $\Rightarrow$ Fill every host bit with "1"

Network bits $\Rightarrow$ 0000

Host bits $\Rightarrow$ 1111

Entire Octet $\Rightarrow$ 00001111

decimal $\Rightarrow$ 15

Broadcast address $\Rightarrow$ 192.168.2.15

7) Last Host:

Rule $\Rightarrow$ Broadcast - 1

192.168.2.14

(0) 192.168.2.14